Poker Hand Classification Using Machine Learning

A Comparative Study of Feature Methods and Models

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CS 488-01: Intro to Big Data Computing

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Big Data Context

5V Model

- Volume: many **combinations**
- Velocity: **real-time** potential
- Variety: mixed **feature types**
- Veracity: **structured data**
- Value: strong **benchmark**

Implication

- **Structure > size**
- **Relationships drive accuracy**

Defining the Problem Space

Problem

- Classify **5-card hands** into **10 categories**
- Features: **suit + rank**
- Values are **categorical**, not numeric

Challenge

- Strong **feature interactions**
- **Nonlinear patterns**
- **Class imbalance**

Hypothesis

If statistical and machine learning techniques analyze card combinations, then they can accurately classify poker hands, because the frequency and structure of card ranks and suits follow consistent probabilistic patterns.

Feature and Modeling Techniques

Data Pipeline

- Train/test split (**80/20**)
- **SMOTE** (train only)
- **Scaling** (fit on train)
- Prevent **data leakage**

Feature Methods

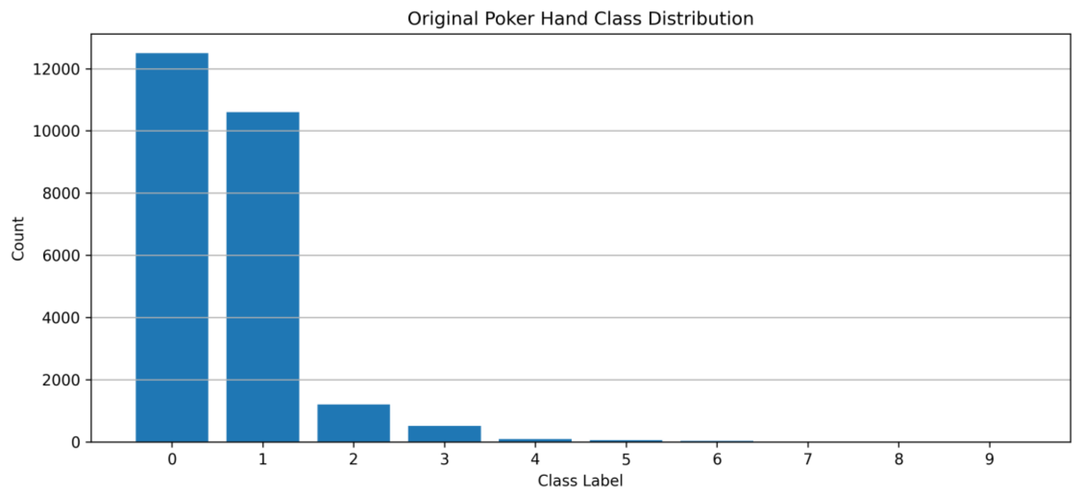
- **Domain features:**
  - Unique rank count
  - Rank range
- **One-hot encoding** → categorical clarity
- **LDA** → dimensional reduction

Model Selection

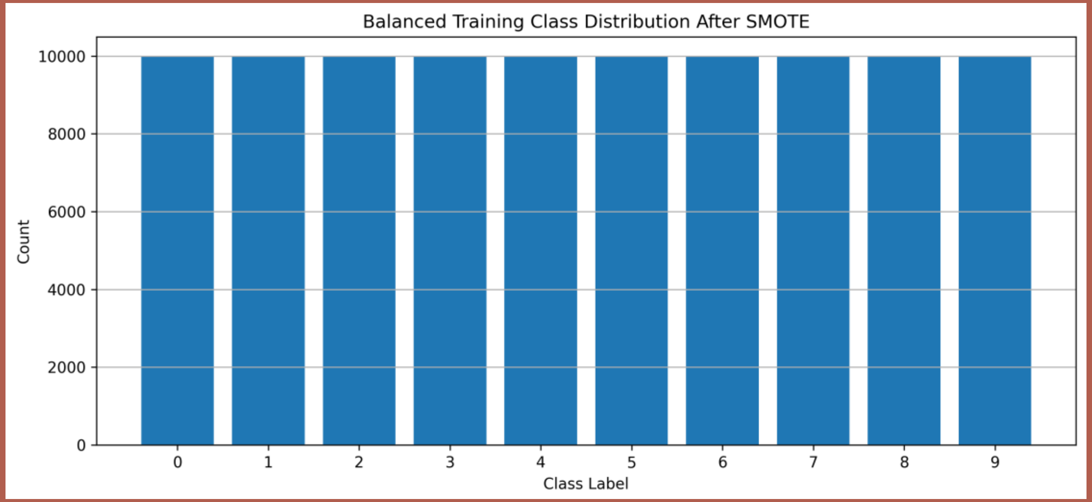
- **Linear SVM**
- **RBF SVM**
- **Random Forest**
- **XGBoost**
- **Nonlinear models** perform better

Class Imbalance

Before

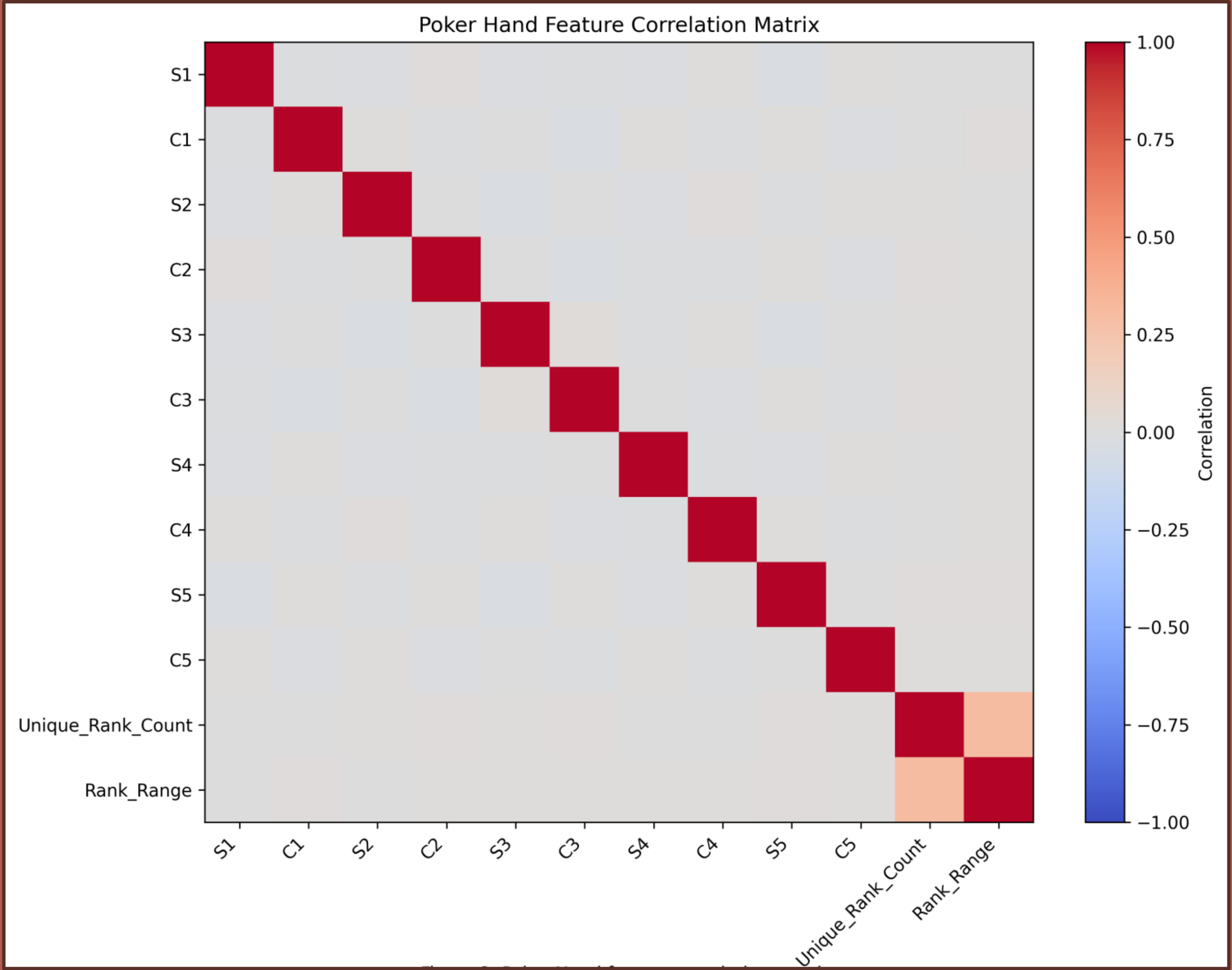


After SMOTE



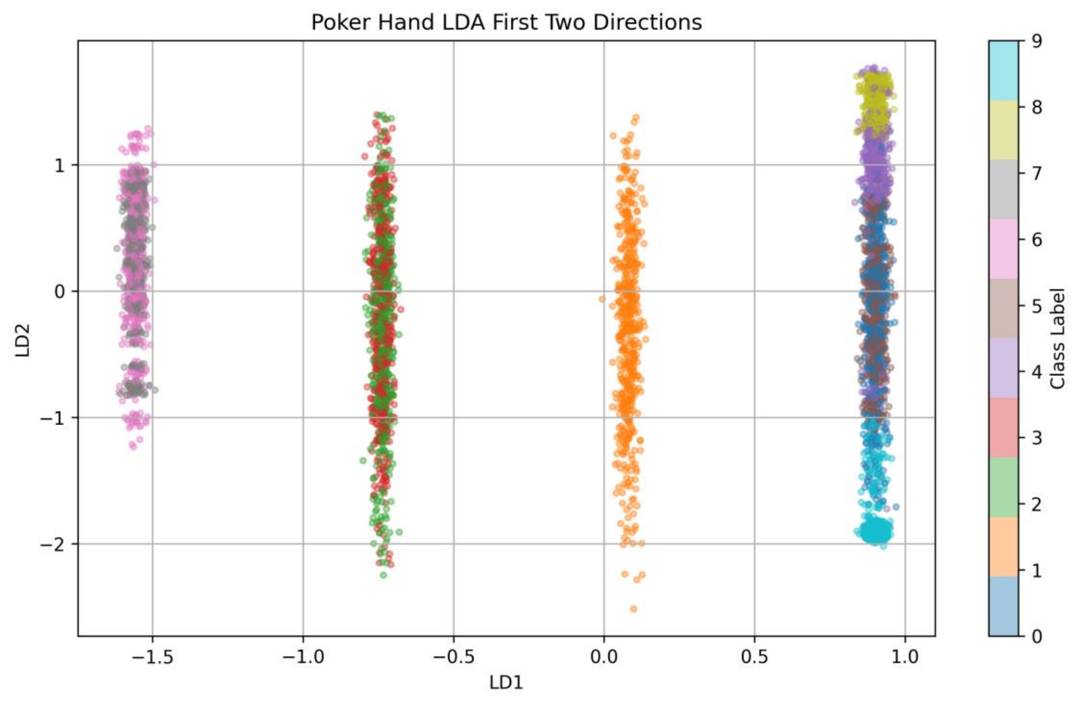
Feature Relationships

- Weak **correlations**
- No dominant feature

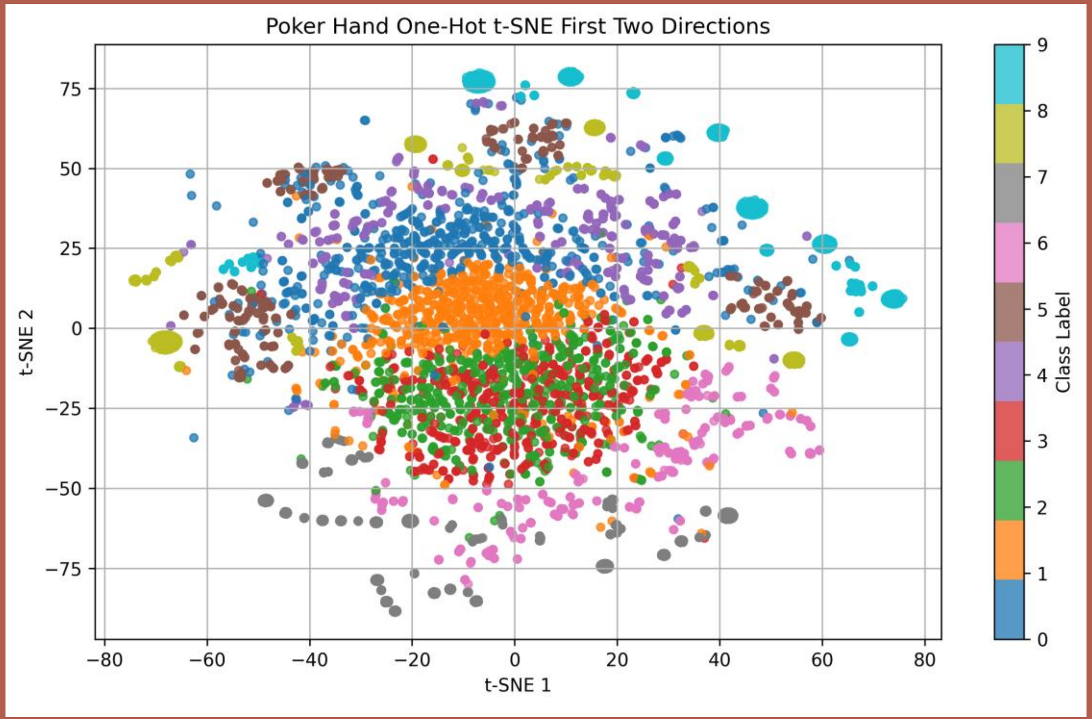


Feature Space Comparison

LDA

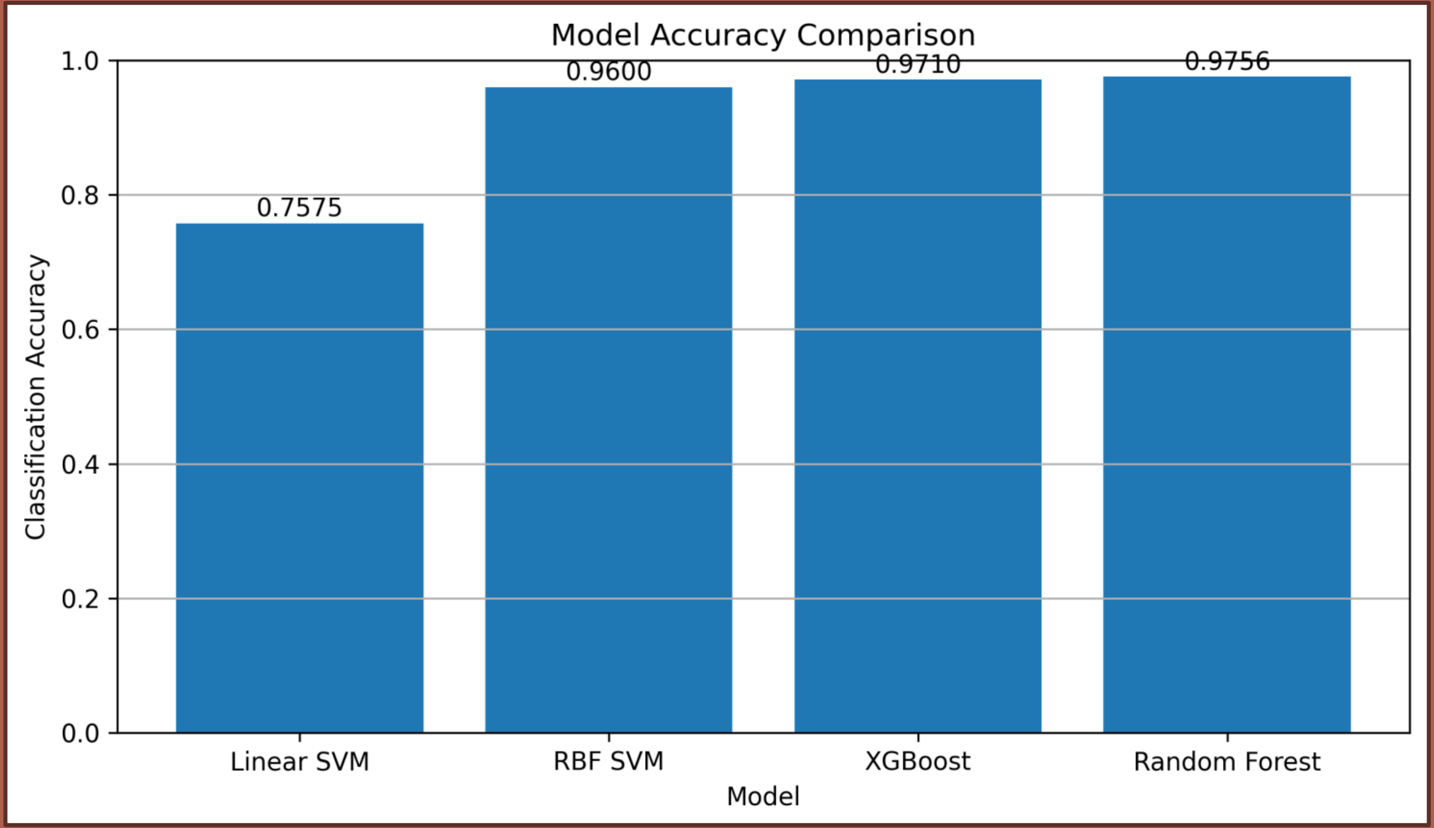


t-SNE



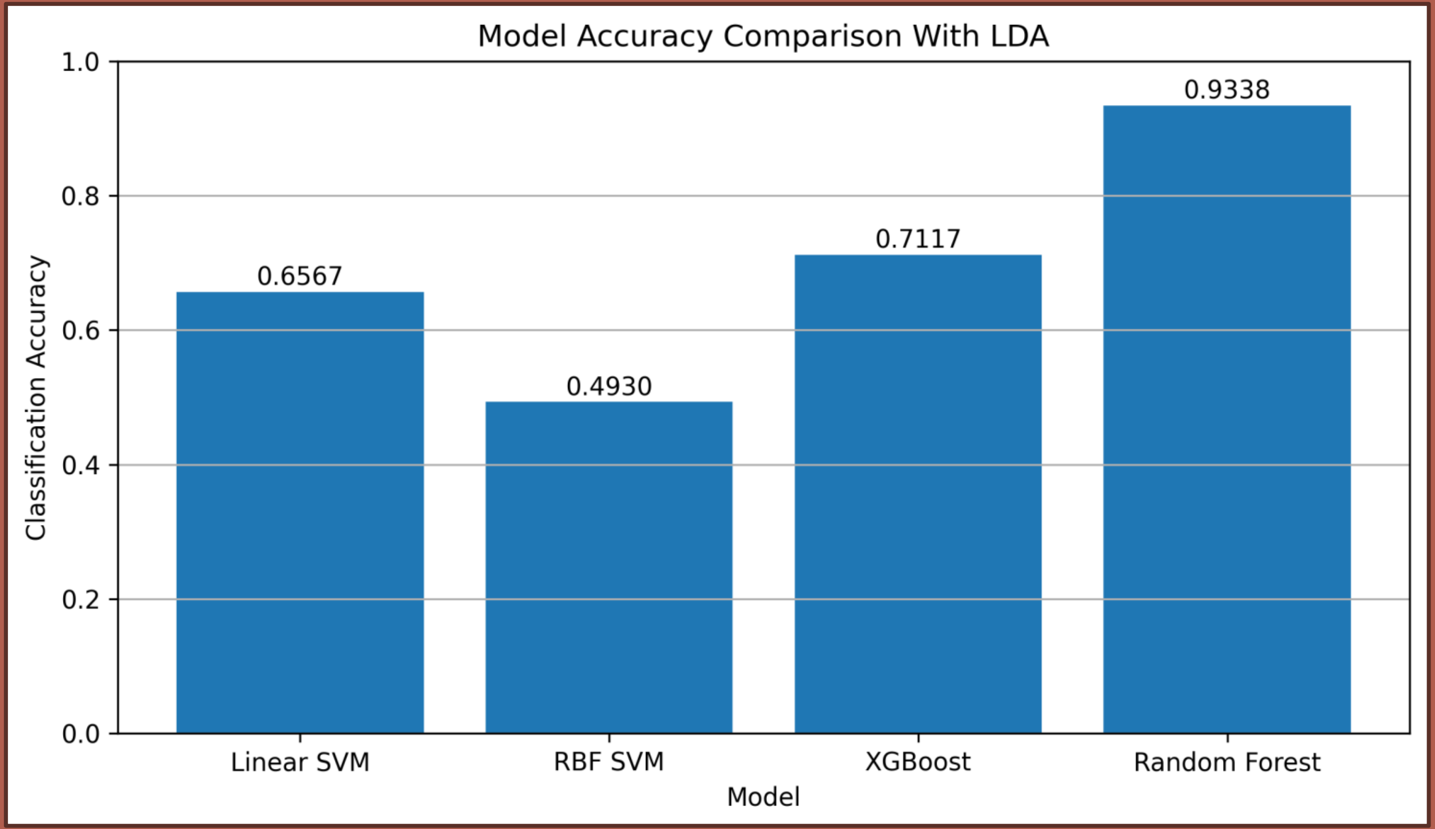
Baseline Performance

- RF ≈ **97.6%** (best)
- XGB, RBF close
- Linear SVM **low**



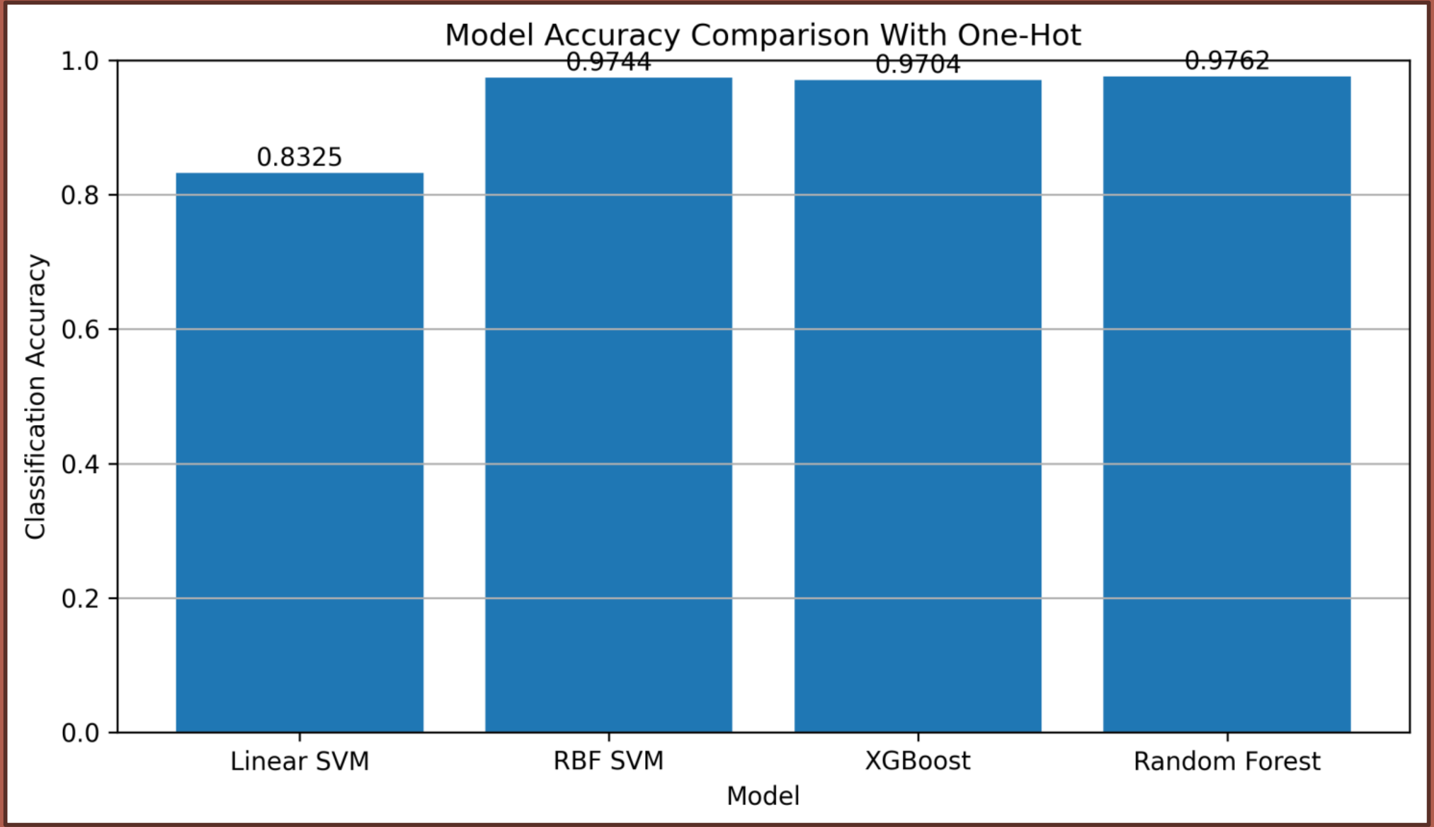
LDA Performance

- Accuracy **drops**
- **Information loss**



One-Hot Performance

- RF **best overall**
- RBF **improves**
- Linear improves slightly



Model Behavior and Key Findings

Performance Patterns

- RF best
- RBF next
- XGB close
- Linear worst
- **Nonlinear > linear**

Core Findings

- **One-hot improves** results
- **LDA reduces** accuracy
- Nonlinear models dominate
- **RF most consistent**

Human vs. AI

Human

- Designed **pipeline**
- Selected **features/models**
- Interpreted **results**

Model

- Learned **patterns**
- Scaled **predictions**

- **Hypothesis supported** — models learned card patterns effectively
- Feature representation + **nonlinear models** → highest accuracy
- Best: **Random Forest + One-Hot**